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Landa C

NIN-Verified. Fraud-Fre

VERSION	DATE	STATUS	OWNER
1.0	March 2026	Draft for Review	Product Team

CORE MISSION

Use NIN-verification to eliminate rental fraud and automate compound bill-s

01

Executive Summary

Landa OS is a next-generation property management platform purpose-built for the Nigerian rental market. Its core mission is to eliminate rental fraud and automate compound-level bill-sharing through mandatory NIN biometric verification, creating a legally binding and digitally accountable rental ecosystem.

THE PROBLEM

- → Widespread rental fraud, impersonation, and identity misrepresentation
- → Disputes over shared utility bills in multi-unit compounds with no transparent allocation
- → Unenforceable tenancy agreements due to lack of biometric authentication
- → Delayed rent payments with no escrow protection for either party
- → No centralized, trusted directory of property artisans and maintenance workers

THE 5 CORE INNOVATIONS

🔐

Tenancy Bridge

Rent payment unlocked ONLY after both parties execute a biometric-signed tenancy agreement.

🧠

Compound Wall

Social accountability dashboard with weighted/equal bill-splitting logic for all compound units.

✅

NIN Verification

Mandatory NIN + Liveness Selfie via Smile ID before any dashboard access is granted.

🏦

Escrow Payments

Rent held 48 hours post-payment via Paystack/Flutterwave — protecting both parties.

🔧

Maintenance Hub

Curated Landa-verified artisan directory with landlord endorsement tags.

⚖️

NDPR Compliant

Zero-storage NIN policy, AES-256 encryption, and full data subject rights

implementation.

BUSINESS MODEL

Revenue Stream	Description	Target
5% Commission	On every rent payment processed through the platform	All Tenants
Basic Plan	Core dashboard + 1 unit management	₦2,500/mo
Growth Plan	Up to 10 units, Compound Wall, Maintenance Hub	₦7,500/mo
Enterprise Plan	Unlimited units, API access, custom branding, analytics	₦20,000/mo
Artisan Listing	Premium placement in Maintenance Hub directory	Artisans/Vendors

YEAR 1 SUCCESS METRICS

10K+

Verified users onboarded

2K+

Active tenancy agreements

₦500M+

Rent processed via escrow

95%

Fraud report reduction

500+

Landa-verified artisans

99.5%

Platform uptime SLA

02

User Personas & User Journeys

PERMISSIONS MATRIX

Permission	Owner	Caretaker	Verified Agent	Tenant
Create/manage properties	✅ Yes	Assigned	Managed	❌ No
Execute tenancy agreements	✅ Yes	Delegated	✅ Yes	✅ Yes
View Compound	✅ Yes	✅ Yes	✅ Yes	Own compound

Wall				
Log cash expenses	✅ Yes	✅ Yes	❌ No	❌ No
Initiate rent payment	❌ No	❌ No	❌ No	✅ Yes
Tag 'Preferred Artisan'	✅ Yes	Delegated	❌ No	❌ No
Manage subscriptions	✅ Yes	❌ No	❌ No	❌ No

PERSONA PROFILES

🏠

Emeka, 52

Owner / Landlord · Lagos Island

PAIN POINTS

Chasing late rent, compound utility disputes, fraudulent tenant applications

GOALS

Streamline rent collection, maintain unit accountability, verify tenants upfront

LANDA OS VALUE

Tenancy Bridge + Compound Wall + NIN verification eliminates all three pain points

👩

Ngozi, 28

Tenant · Abuja

PAIN POINTS

Inflated utility bills, no proof of payment, unsafe cash transactions

GOALS

Transparent bill allocation, digital receipts, safe rental experience

LANDA OS VALUE

Compound Wall transparency + Escrow protection + Digital tenancy agreement



Abdul, 38

Caretaker · Kano

PAIN POINTS

Managing multiple units manually, disputes when landlord absent

GOALS

Log expenses easily, communicate payment statuses, manage tenant queries

LANDA OS VALUE

Delegated permissions, photo expense logger, mobile-first UX



Chidi, 35

Verified Agent · Port Harcourt

PAIN POINTS

Difficulty proving legitimacy, slow agreement execution, commission disputes

GOALS

Close deals faster, build verified reputation, manage multiple clients

LANDA OS VALUE

'Verified Agent' badge + ability to manage assigned properties

JOURNEY 1: TENANT ONBOARDING & FIRST RENT PAYMENT

1 **Download & Register**

Tenant enters phone + email; OTP verification sent and confirmed.

2 **NIN + Liveness Selfie**

Smile ID API validates NIN and performs anti-spoofing liveness check.

MUST PASS

3 **Status: Verified**

Dashboard unlocked. 'Verified' badge applied to profile.

4 **Receive Agreement Link**

Landlord sends tenancy agreement to tenant's Landa OS inbox.

5 **Tenant Signs Biometrically**

Biometric re-confirmation via Smile ID; blockchain-timestamped signature recorded.

MUST SIGN

6 **Landlord Signs Biometrically**

Tenancy Bridge fully activated. Both signatures immutably recorded.

MUST SIGN

7 **Pay Rent Button Unlocked**

Paystack/Flutterwave payment initiated with 5% platform fee shown transparently.

8 **48-Hour Escrow Hold**

Funds held; landlord notified; dispute window open for tenant.

9 **Escrow Released**

Funds auto-released to landlord after 48h. Digital receipts issued to both parties.

03

Functional Requirements

FR-AUTH-001 **NIN + Liveness Selfie Onboarding**

- All users must complete NIN verification before accessing any dashboard feature
- Integration: Smile ID API — NIN lookup + liveness selfie with anti-spoofing
- Status transitions: Unverified → Pending → Verified | Failed
- Failed verification: User receives reason code; up to 3 attempts before 24-hour cooldown
- 'Verified' badge displayed on all user profiles and tenancy agreements

FR-AUTH-002 **Access Control (RBAC)**

- → Dashboards and payment flows are completely inaccessible until 'Verified' status is achieved
- → Role-based access control enforced at the API layer — frontend gatekeeping is supplementary only
- → Session tokens expire after 30 minutes of inactivity; biometric re-auth required for payments

FR-TB-001 **The Tenancy Bridge — Agreement Execution**

- → Landlord/Agent creates agreement from standardized template (customizable: unit, duration, rent, utilities, rules)
- → Agreement sent to tenant's Landa OS inbox — NOT external email — for full traceability
- → Both parties must biometrically sign for agreement to be 'Executed'
- → Agreement is timestamped, hash-signed, and stored as immutable record (Section 84 Evidence Act compliant)
- → Only AFTER both signatures: 'Pay Rent' button is unlocked for the tenant

FR-MF-001 **Rent Payment Processing**

- Primary gateway: Paystack | Fallback: Flutterwave
- Supported methods: Card, Bank Transfer, USSD, Paystack Balance
- Platform charges 5% commission; displayed transparently at checkout
- Tenant sees: Rent Amount + 5% Fee = Total | Landlord receives: Rent Amount - 5% Fee

FR-MF-002 **Escrow Logic**

- All rent held in escrow for exactly 48 hours post-payment confirmation
- During 48-hour window: Tenant can raise a dispute; triggers manual review and escrow freeze
- No dispute after 48 hours: Funds auto-released to landlord's wallet or linked bank account
- Escrow balance and release schedule visible to both parties in real time

FR-CW-001 **Compound Wall — Bill Splitting Engine**

- Owner selects split method: Equal Split OR Weighted Split (by occupants, sqm, or custom %)
- System calculates per-unit allocation; rounds to nearest ₦1; remainder goes to largest unit
- Tenants receive push notification + in-app bill detail showing exact formula used

FR-CW-002 **Compound Wall — Social Accountability UI**

- Grid of all units with color-coded status: 🟢 Paid · 🟡 Due Soon · 🔴 Overdue · ⚪ No Agreement
- Unit identifiers visible to all compound residents — intentional social accountability design
- Tenant names NOT shown to other tenants (privacy-preserving); only unit identifier and status
- Owner/Caretaker sees full tenant names and contact options

FR-CW-003 **Cash Expense Logger**

- Landlord/Caretaker logs expenses with: Category, Amount (₦), Date, Description, Photo receipt (mandatory)
- Logged expenses visible to all tenants in the compound for transparency and manual verification
- Tenants can 'Acknowledge' or 'Dispute' any logged expense; disputes flagged to Landa OS support
- Expense history exportable as PDF for accounting purposes

FR-MH-001 **Maintenance Hub — Artisan Directory**

- Open directory of trades: Plumbers, Electricians, Carpenters, Painters, Cleaners, HVAC, etc.
- All artisans must be 'Landa-Verified': NIN check + address confirmation + at least one landlord endorsement
- Profiles show: Trade, Service areas, Landa-Verified badge, Rating, 'Landlord Preferred' tags
- Landlords can tag an artisan as 'Preferred' for their specific compound/unit

04 Database Schema (ERD)

Implementation uses PostgreSQL with row-level security (RLS) enforcing RBAC at the database layer. Raw NIN numbers are NEVER stored — only Smile ID tokens.

users

`id`	UUID primary key
`phone, email`	Encrypted PII fields
`role`	ENUM: Owner | Caretaker | Agent | Tenant
`nin_token`	Smile ID token ONLY — raw NIN never stored
`verification_status`	ENUM: Unverified | Pending | Verified | Failed
`smile_id_ref`	Reference to Smile ID verification record

tenancy_agreements

`id`	UUID primary key
`unit_id, tenant_id`	Foreign keys
`landlord_id`	FK → users
`status`	ENUM: Draft | Pending | Executed | Expired | Voided
`tenant_signature_hash`	SHA-256 biometric signature hash
`signed_at`	NTP-synchronized timestamp

payments

`id`	UUID primary key
`agreement_id`	FK → tenancy_agreements
`amount, platform_fee`	Gross amount and 5% commission
`paystack_ref`	Paystack transaction reference
`status`	ENUM: Pending | Escrow | Released | Disputed | Refunded

escrow_release_at	Drives 48-hour auto-release job

compound_bills

id	UUID primary key
property_id	FK → properties
bill_type	Electricity, Water, Waste, etc.
total_amount	Total bill in ₦
split_method	ENUM: Equal | Weighted
receipt_photo_url	Encrypted S3/R2 URL

artisans

id	UUID primary key
user_id	FK → users (NIN verified)
trade_category	Plumber, Electrician, Carpenter, etc.
coverage_areas	JSONB — city/LGA coverage list
is_landa_verified	Boolean; set by Landa admin
rating_avg	Aggregated from landlord endorsements

audit_log

id	UUID primary key
user_id	Actor performing action
action	Descriptive action string
entity_type, entity_id	Polymorphic target reference
ip_address	Request IP (NDPR logging)
timestamp	Immutable append-only; no UPDATE allowed

KEY RELATIONSHIPS

- → users (1) → (many) properties [via owner_id] → (many) units
- → units (1) → (many) tenancy_agreements [one ACTIVE at a time; history preserved]
- → tenancy_agreements (1) → (many) payments [monthly rent cycles]
- → properties (1) → (many) compound_bills → (many) unit_bill_allocations
- → artisans (1) → (many) artisan_endorsements from different landlords

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05

API Integration List



Smile ID API

NIN verification + Liveness Selfie anti-spoofing. Endpoints: POST /v1/id_verification (NIN lookup), POST /v1/smile_links (liveness), Webhook: result callback

Identity



Paystack (Primary)

Rent payments, subscriptions, payouts. POST /transaction/initialize, GET /transaction/verify/{ref}, POST /subscription/create, Webhook: charge.success

Payments



Flutterwave (Fallback)

Fallback payment processor. POST /payments, GET /transactions/{id}/verify, POST /refunds, Webhook: charge.completed

Payments

Google Maps Platform

Property geolocation, artisan coverage mapping, address autocomplete. Places API, Geocoding API, Maps JavaScript SDK

Location

Firebase Cloud Messaging

Push notifications for payment alerts, bill splits, agreement status changes. Firebase Admin SDK

Notifications

AWS S3 / Cloudflare R2

Storage for photo receipts, tenancy agreement PDFs, artisan verification photos. AES-256 encrypted at rest.

Storage

SendGrid / Termii

Transactional email + SMS for OTP delivery, payment receipts, and renewal reminders

Messaging

Mono / Okra (Optional)

Open Banking: bank account verification for landlord payouts. POST /accounts/lookup, GET /transactions

Banking

06

Security & Compliance Roadmap

NDPR COMPLIANCE — DATA MINIMIZATION

- → Raw NIN numbers are NEVER stored — only the tokenized Smile ID reference is retained (Zero-Storage Policy)
- → Liveness selfie frames processed by Smile ID and NOT stored by Landa OS beyond the verification session
- → All personal data collection requires explicit consent linked to the Privacy Policy (NDPR Art. 2.1)
- → Data retention: Active data kept for account duration + 7 years (financial); deleted within 30 days of closure request

DATA SUBJECT RIGHTS (NDPR ART. 3.1)

Right	Implementation	SLA
Right to Access	Users download their data from in-app settings	72 hours
Right to Erasure	Non-financial data deleted on request; financial records per CBN (7 years)	30 days
Right to Portability	Tenancy agreements and payment history exportable as PDF/CSV	Immediate
DPO	Data Protection Officer designated; contact	—

Contact	published in-app and on website	

ENCRYPTION STANDARDS

At Rest

AES-256 column-level encryption for all PII (phone, email, bank refs). S3/R2 bucket-level encryption (SSE-KMS). Database backups encrypted with separate KMS keys in geographically separate region.

In Transit

TLS 1.3 minimum on all endpoints; TLS 1.0/1.1 explicitly disabled. Certificate pinning in mobile clients to prevent MITM. HMAC signature verification on all webhooks (Paystack/Flutterwave standard).

SECTION 84 EVIDENCE ACT COMPLIANCE

- → All agreements stored with: SHA-256 hash, biometric signature reference, NTP-synchronized timestamp, immutable audit log entry
- → Agreements generated in PDF/A format (archival standard) for long-term admissibility
- → No agreement record can be modified after execution — updates require a new agreement with version reference

SECURITY ROADMAP

PHASE 1

Foundation

Pre-Launch (Month 1–2)

- TLS 1.3 enforcement
- AES-256 at rest
- RBAC database layer
- Smile ID integration
- NDPR Privacy Policy published

PHASE 2

Hardening

Month 3–4 Post-Launch

- Penetration test (VAPT)
- Bug bounty program
- Automated dependency scanning
- SOC 2 Type I prep

PHASE 3

Certification

Month 6–12

- ISO 27001 gap analysis
- NDPR registration (NITDA)
- CBN PSP license application
- SOC 2 Type II audit

PHASE 4

Continuous

Ongoing (Annual)

- Quarterly VAPT
- Annual DPO report
- Incident response drills
- NDPR compliance review

07 Non-Functional Requirements

Category	Requirement	Target
Performance	API response time (95th percentile)	< 400ms
Performance	Payment processing end-to-end	< 8 seconds
Performance	Smile ID verification response	< 15 seconds
Availability	Platform uptime SLA	99.5%
Availability	Payment processing uptime	99.9%
Scalability	Concurrent active users at launch	5,000+
Scalability	Peak payment	2,000+

	transactions/hour	
Mobile	App download size (iOS + Android)	< 30MB
Mobile	Offline capability	View last-synced dashboard; queue actions
Accessibility	Language support	English + Pidgin English UI labels
AML/KYC	Enhanced due diligence threshold	Transactions above ₦500,000

08 Glossary

Term	Definition
NIN	National Identification Number — Nigeria's 11-digit unique biometric identifier issued by NIMC
Smile ID	African identity verification API providing NIN lookup + liveness selfie anti-spoofing
Tenancy Bridge	Feature blocking rent payment until both parties execute a biometric-signed tenancy agreement
Compound Wall	Social accountability dashboard showing payment status for all units with bill-splitting logic
Escrow	48-hour payment hold mechanism protecting both tenant and landlord during dispute window

Landa-Verified	Status for artisans who pass NIN check + address verification + at least one landlord endorsement
NDPR	Nigeria Data Protection Regulation — enforced by NITDA
RBAC	Role-Based Access Control — permissions assigned by role (Owner, Caretaker, Agent, Tenant)
Section 84 EA	Section 84 of the Nigerian Evidence Act 2011 governing admissibility of computer-generated evidence
CBN	Central Bank of Nigeria — regulatory authority for payment service providers
NITDA	National Information Technology Development Agency — regulates NDPR compliance
Liveness Selfie	AI-powered biometric check confirming a real person (not a photo/video) is performing verification
Weighted Split	Bill allocation proportional to unit occupancy, size, or custom percentage
DPO	Data Protection Officer — responsible for NDPR compliance oversight

This document is intended solely for authorized Landa OS stakeholders.